

International Comparisons of Living Standards by Equivalent Incomes*

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Abstract

We propose a measure of living standards for international comparisons. Based on GDP per capita, the measure incorporates corrections for international flows of income, labor, risk of unemployment, healthy life expectancy, household demography and inequalities. The method for comparing populations that differ in some non-income dimension consists of computing the equivalent variation of income that would make each population indifferent between its current situation and a reference situation with respect to the non-income dimension. This is applied to 24 OECD countries. The obtained ranking of countries differs substantially from the GDP ranking.

Keywords: Social welfare; money-metric utility; GDP; OECD

JEL classification: D60; D71; O57; P17

I. Introduction

International comparisons of living standards are still primarily made with GDP per capita, in spite of recurrent criticism that this is a partial and ill-founded measure of social welfare; see e.g. Sen (1979). Alternative measures abound, such as the Index of Human Development computed by the United Nations Development Program since 1990, Osberg and Sharpe's (2002) Index of Economic Well-being, and Miringoff and Miringoff's (1999) Index of Social Health. They are based on the aggregation of various sub-indices of social performance, and the weights used for this aggregation have no rational basis.

In this paper, we propose to rely on a basic notion of welfare economics, namely, equivalent income (also called money-metric utility), in

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order to make international comparisons that take account of inter-country differences in non-income dimensions. Specifically, when countries differ in some non-income dimension (such as leisure or health, for instance), we fix a reference level for this dimension and, for each country, compute the willingness to pay (WTP) of the population in order to obtain this reference level. Correcting current income by this amount, we obtain an "equivalent income". This is directly comparable across countries because it corresponds to a level of income that, when combined with the reference level for the non-income dimension, gives a situation that is, in the eyes of the population, equivalent to their current situation. In a nutshell, we reduce all differences to income differences so as to make comparisons possible in the monetary dimension.

Social choice theorists have often been critical of monetary measures of well-being. But we argue that the equivalent income can be given a sound basis in social choice theory. We hope to bridge the gap between this theory and applied studies, suggesting how applications can be imagined along this vein. Our work is also related to Becker, Philipson and Soares (2005), who compute equivalent growth rates that incorporate the monetary equivalent of improved life expectancy. Their analysis, however, is limited to the study of growth rates, whereas we examine how to compare levels of standards of living. In addition, we take account of inequalities in the distribution of income so as to avoid counting a dollar for the poor as equivalent to a dollar for the rich.

We apply this approach to a sample of 24 OECD countries for the year 2004. The non-income dimensions that are incorporated into the computations involve leisure, health, risk of unemployment and household composition. Table 1 compares our results for a sub-sample of countries with the ranking obtained with GDP per capita and the HDI. Indices are expressed as percentages of the US index. As is well known, the HDI suppresses the differences between rich countries because it is based on a rescaling that takes account of all rich and poor countries. In contrast, our indicator displays much more variability. The ordinal ranking of countries is also different. For instance, Australia fares well with the HDI because of good health and education sub-indices. In contrast, it fares less well with our indicator because of a low level of leisure and a high level of inequalities (and also a national income much below GDP). Sweden also fares less well with our indicator, but because of small household size. For Finland, the two factors of leisure and household size combine their negative effect. In contrast, some countries are better treated by our indicator than by the HDI, such as Denmark, France, Germany and the Netherlands, mainly thanks to the leisure dimension. The table also shows that more generally, the corrections are far from negligible with our indicator. Not surprisingly, the final index of living standard is still correlated with GDP per

Table 1. *Illustration of result (2004)*

	GDP per capita		HDI		Our indicator	
	Percentage	Rank	Percentage	Rank	Percentage	Rank
Australia	76	8	100.9	2	82	11
Canada	79	6	100.2	5	86	9
Denmark	81	5	99.5	10	89	7
Finland	75	9	99.9	8	80	15
France	73	12	99.4	11	93	6
Germany	71	14	98.3	15	81	13
Ireland	101	1	100.8	3	115	2
Italy	71	13	99.2	12	88	8
Japan	75	10	100.1	6	101	3
Netherlands	81	4	99.9	8	94	5
Norway	97	3	101.8	1	117	1
Spain	64	15	98.9	14	80	14
Sweden	74	11	100.3	4	81	12
United Kingdom	78	7	99.2	12	84	10
United States	100	2	100.0	7	100	4

Sources: OECD; *World Development Report 2006*; authors' computations.

capita, but the general ranking of countries is substantially affected by the corrections.

The next section explains the social choice theoretical underpinnings of our approach. Section III introduces the model and explains how our computations are made. Section IV presents the results about living standards for our sample of 24 OECD countries. The last section concludes. Information about the data used in the computations is provided in the Appendix. An important feature of our work is that we have made the most of easily accessible data, and have not sought to dig further into heterogeneous national statistics at the risk of making the results harder to reproduce. We propose this work as a prototype that could encourage other researchers to improve on it and could encourage producers of statistics to refine their data in this direction. In the conclusion, we describe the new kind of international surveys on consumption and preferences that are needed in order to apply the theory in a more accurate way.

II. Equivalent Income

The purpose of this section is to introduce and justify the measurement of social welfare in terms of equivalent incomes. We first present the notions and then relate the approach to relevant branches of welfare economics. The last sub-section introduces some approximation methods for the computation of social welfare.

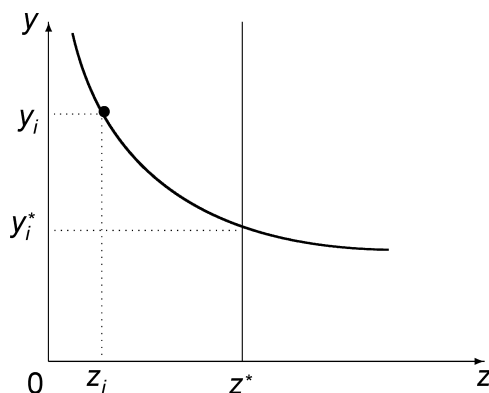


Fig. 1. Equivalent income

The Notion of Equivalent Income

This section relies on a simple static framework that is sufficient for the discussion. Consider a population of individuals $i = 1, \dots, n$ with monetary incomes y_i who face living conditions described by a vector z_i . The components of z_i may include market prices and wages, quantities of public goods and externalities, personal situations with respect to family and health, and so on. Individuals have an indirect utility function $v_i(y_i, z_i)$, which is equal to the maximal utility they can achieve after adjusting their demand for commodities. We assume v_i to be increasing in y_i .

Fix a benchmark z^* and compute the solution y_i^* to the equation $v_i(y_i, z_i) = v_i(y_i^*, z^*)$. The quantity y_i^* may be called i 's "equivalent income". It is illustrated in Figure 1. Under the assumption that v_i is increasing in y_i , the equivalent income correctly represents individual preferences since a better situation for i necessarily corresponds to a greater equivalent income:

$$v_i(y_i^*, z^*) = v_i(y_i, z_i) > v_i(y_i', z_i') = v_i(y_i'^*, z^*) \Leftrightarrow y_i^* > y_i'^*. \quad (1)$$

In order to evaluate the situation of the whole population, we propose applying an inequality-averse social welfare function W to the vector of equivalent incomes $(y_1^*, \dots, y_n^*)$. Social welfare is then $W(y_1^*, \dots, y_n^*)$. The next sub-section explains the connection with various branches of welfare economics and examines the pros and cons of this approach.

Relation to Various Branches of Welfare Economics and Discussion of the Approach

This sub-section explains how our approach relates to money-metric utility, cost-benefit analysis, social choice theory, equivalence scales and pricing.

Money-metric Utility. In the particular case when z_i describes the market prices, equivalent income has been introduced by Samuelson (1974) and Samuelson and Swamy (1974) under the denomination “money-metric utility”. Their purpose was to find an index number that represented individual preferences while being expressed in quantity (or monetary) units. The idea of applying a social welfare function to the distribution of equivalent incomes was later put forth by various authors, in particular King (1983), who introduced the expression “equivalent income”. We propose a simple generalization of this idea to the case in which z_i encompasses all relevant non-monetary dimensions of life.¹

Money-metric utilities have attracted criticism that may also be directed at our approach. A first criticism is that a social welfare function $W(y_1^*, \dots, y_n^*)$ may fail to be quasi-concave in commodities; see Blackorby and Donaldson (1988). This observation has been generalized in the theory of fair allocation; see e.g. Maniquet and Sprumont (2004): any approach (i.e., not just money-metric utilities) that obeys the Pareto principle and evaluates individual situations by the indifference sets at the contemplated allocation will fail to be quasi-concave unless it gives absolute priority to the worst off. The literature on fair allocation concludes from this result that only criteria of the maximin or leximin sort are acceptable. We adopt here a more pragmatic attitude based on the observation that violations of quasi-concavity are unlikely to be severe for well-behaved functions.

A second criticism, due to Roberts (1980) and Slesnick (1991), targets the fact that money-metric utility functions depend on a reference price vector, and that if one wants the social welfare function to be independent of these reference parameters severe restrictions are required (e.g. homothetic preferences). Similarly, the computation of equivalent incomes in our approach is sensitive to the choice of the reference z^* . But we claim that the choice of z^* need not be arbitrary, as has been shown recently in the literature on fair allocation. In particular, z^* has the special property that when individuals enjoy $z_i = z^*$, their situations can be compared in terms of ordinary incomes y_i , independent of their preferences. This property may help in selecting z^* . For instance, concerning health, it is standard to ignore preferences about health and consumption when comparing individuals in good health and to take account of preferences about health and consumption when comparing individuals in bad health. This implies that the reference level of health in z^* should be good health.²

Another criticism raised by Sen (1979) is that money-metric utilities neglect welfare inequalities having to do with special needs. Needs are,

¹ A similar generalization was already studied by Hammond (1994) in relation to the cost-benefit analysis of environmental externalities.

² A more rigorous version of this argument can be found in Fleurbaey (2005).

however, easily incorporated into the equivalent income approach because they may feature in the dimensions of z_i .

Cost-Benefit Analysis. When the situation of i changes from (y_i^0, z_i^0) to (y_i^1, z_i^1) , the compensating variation is computed as the quantity CV_i such that $v_i(y_i^0, z_i^0) = v_i(y_i^1 - CV_i, z_i^1)$. In other words, it corresponds to what i would be willing to pay in order to incur the change. Cost-benefit analysis traditionally evaluates a change for the whole population by the sign of $\sum_{i=1}^n CV_i$. When $z_i^1 = z^*$, this is equivalent to checking the sign of $\sum_{i=1}^n (y_i^1 - y_i^*)$.³

Social choice theorists have raised two main criticisms against this practice.⁴ First, there is no guarantee that it produces a consistent ranking of all possible situations, and examples of intransitive rankings have been exhibited. Second, it is utterly questionable to add the WTP (willingness to pay) of the rich and the poor, as if a dollar were worth the same for everyone. Our approach avoids the two problems, first by taking a fixed reference z^* (instead of the moving, personal reference z_i^1), and second by using an inequality-averse social welfare function (instead of the sum).

Social Choice Theory. The equivalent income is computed from individual ordinal preferences and does not depend on additional information about utility functions. It is sometimes alleged that, in virtue of Arrow's impossibility theorem (Arrow, 1951), there is no satisfactory ranking of social situations on the sole basis of individual ordinal non-comparable preferences. Arrow's theorem, however, involves the restrictive condition of independence of irrelevant alternatives (IIA), which says that the ranking of two situations should be based only on individual preferences about these two situations exclusively. Our approach characteristically violates this restriction, since the ranking of two situations may depend on how individuals compare them to other situations involving the reference z^* .

The question is therefore to assess whether IIA is compelling. First, one can observe that traditional cost-benefit analysis also violates this condition, since the evaluation of a change from (y_i^0, z_i^0) to (y_i^1, z_i^1) involves the consideration of a third alternative $(y_i^1 - CV_i, z_i^1)$. To our knowledge, cost-benefit analysis has never been criticized for violating IIA. On the contrary, many authors have noted that IIA is very restrictive since it precludes even

³ Using equivalent variations, defined by

$$v_i(y_i^0 + EV_i, z_i^0) = v_i(y_i^1, z_i^1),$$

amounts to taking $z^* = z_i^0$ and checking the sign of $\sum_{i=1}^n (y_i^* - y_i^0)$. Becker *et al.* (2005) compute the equivalent growth rate $\sum_{i=1}^n (y_i^* - y_i^0) / \sum_{i=1}^n y_i^0$.

⁴ See e.g. Arrow (1951), Boadway and Bruce (1984), Blackorby and Donaldson (1990).

the reference to marginal rates of substitution.⁵ Therefore, we conclude that Arrow's theorem cannot be used as an objection against our approach.

The closely related theory of fair allocation has developed the concept of "egalitarian-equivalent" allocations. These were initially defined by the fact that all individuals are indifferent between their consumption bundle and a reference bundle (Pazner and Schmeidler, 1978), and they were later generalized in terms of indifference to a reference choice set (Thomson, 1994). This notion has also been extended from the definition of fair allocations to the ranking of all arbitrary allocations by applying a social welfare function to the individual indices (Pazner, 1979).⁶ Our approach is simply a special application of this general "equivalence" approach.

Equivalence Scales. The notion of equivalent income has been widely used in the context of evaluating income distributions for households of different types. Traditionally, a household utility function bearing on household income and household type was assumed to make it possible to compute an equivalent income for each household, with the single person as reference type. This approach has raised concerns due to the impossibility of estimating such utility functions from demand data (Pollak and Wales, 1979), and the unitary model of the household is now in disrepute.

We prefer to compute the equivalent income at the level of individuals, not households. One can compute, for each individual, how much income would give him his current utility if he were to live alone. This approach has been recently advocated by Browning, Chiappori and Lewbel (2006).

Pricing. An alternative method that is often considered for the incorporation of living conditions into a monetary measure of living standards consists of using a price instead of a WTP. When there is a market in which individuals choose their personal consumption for the dimension of living conditions under consideration, the *marginal* WTP can indeed be estimated by the price they face (at least in an interior allocation). We will actually rely on this method when computing the correction for leisure time, because the difference between actual and reference leisure time will be small in our figures.

When the correction to be made is more than marginal, this estimation is of course impractical, although the method is unfortunately often applied. It is, for instance, often considered a natural correction for leisure time to add to ordinary income the value of leisure, at current wages, in order to obtain a value of "full income," which is interpreted as the rent value

⁵ An analysis of how much information is needed about individual preferences in order to escape Arrow's impossibility is made in Fleurbaey, Suzumura and Tadenuma (2005).

⁶ This idea was also put forth in Samuelson (1977), and it can be directly connected to the Bergson-Samuelson concept of social welfare. See Fleurbaey and Mongin (2005).

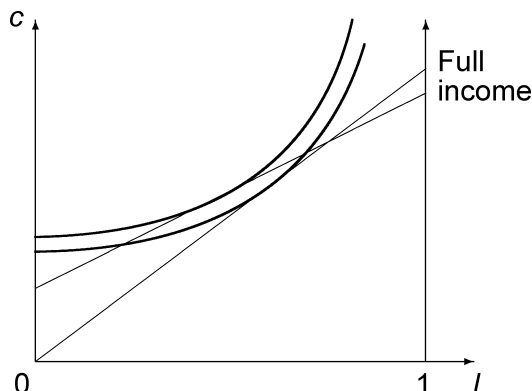


Fig. 2. Full income and preferences

of human capital.⁷ This approach is legitimate if one uses it in order to represent individual preferences. If the wage rate is fixed, the higher the full income, the more satisfied is the individual who can freely choose his labor time, because his budget set has expanded. But this approach is very questionable when used for interpersonal comparisons of living standards, for the following reason. Let us take it as axiomatic that for any approach to the measurement of living standards to be reasonable, it must be such that when two individuals have identical preferences, how these individuals (unanimously) compare their situations yields the correct ranking of their living standards. The full-income measure characteristically fails to satisfy this requirement. When two individuals have identical preferences but different wage rates, the one with the greater full income may well be on the lower indifference curve, as illustrated in Figure 2 (where c denotes consumption and ℓ labor). Indeed, different wage rates make it possible for the budget lines to cross, so that the individual with the greater full income may actually choose a portion of his budget that is dominated by the other budget.⁸

For non-marginal corrections, the pricing method is legitimate only in a special circumstance, namely, when all individuals face the same price and choose freely on a budget set defined with this price. In such a case, the pricing method amounts to comparing individual living standards on the basis of the value of individual endowments at the prevailing market price. In this special context, individuals with identical preferences will indeed always rank their situations in line with the value of their endowments.

⁷ See e.g. Nordhaus and Tobin (1973).

⁸ In contrast, as can be easily seen in Figure 1, two individuals with identical preferences necessarily rank their situations in the same way as their equivalent income, because their indifference curves do not cross.

In summary, the pricing method is acceptable when used as an estimation of marginal WTP, for small corrections, or when all individuals face the same prices and can freely choose in the corresponding budget set. The fact that we take income as the first step in the measurement of living standards, instead of a full consumption vector, is actually based on this idea. But the pricing method should be confined to these specific contexts and should not be extrapolated to other contexts.

From Individual to Social Welfare

From theory to application, we face obstacles that force us to make approximations. Data limitations prevent us from computing y_i^* at the individual level. Instead we first compute an approximate value of the average, $(1/n) \sum_{i=1}^n y_i^*$, in the following way. Let δ_i denote individual i 's WTP to move from z_i to z^* . By definition, $y_i^* = y_i - \delta_i$ and therefore

$$\frac{1}{n} \sum_{i=1}^n y_i^* = \frac{1}{n} \sum_{i=1}^n y_i - \frac{1}{n} \sum_{i=1}^n \delta_i. \quad (2)$$

Willingness to pay, at the individual level, can be approximated by the sum of the WTP for each component k of the vector z :

$$\delta_i \approx \sum_k \gamma_{ik} (z_k^* - z_{ik}), \quad (3)$$

where γ_{ik} is i 's marginal WTP for a unit improvement in z_{ik} .

As a consequence,

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n \delta_i &\approx \frac{1}{n} \sum_{i=1}^n \sum_k \gamma_{ik} (z_k^* - z_{ik}) \\ &= \sum_k \left(\frac{1}{n} \sum_{i=1}^n \gamma_{ik} \right) \left(z_k^* - \frac{(1/n) \sum_{i=1}^n \gamma_{ik} z_{ik}}{(1/n) \sum_{i=1}^n \gamma_{ik}} \right). \end{aligned} \quad (4)$$

If, for every k , the correlation between γ_{ik} and z_{ik} over the population is not too strong,⁹ this can be approximated by

$$\frac{1}{n} \sum_{i=1}^n \delta_i \approx \sum_k \left(\frac{1}{n} \sum_{i=1}^n \gamma_{ik} \right) \left(z_k^* - \frac{1}{n} \sum_{i=1}^n z_{ik} \right). \quad (5)$$

⁹ One might object to this approximation by arguing that a negative correlation between γ_{ik} and z_{ik} is to be expected (the greater z , the lower the WTP for an additional unit), but the correlation at stake here is over the whole population, not over a single indifference curve. For instance, in a competitive market all agents have the same marginal rate of substitution even when they have different levels of consumption. More generally, if there is a positive correlation between individual situations with respect to the various dimensions (k) of z , the correlation between γ_{ik} and z_{ik} has no reason to be negative rather than positive.

In other words, we approximate the average WTP by applying an average marginal WTP to the average situation of the population. This amounts to reasoning in terms of a representative agent, which we will do in this paper. The details about the preferences of the representative agent are presented in the next section.¹⁰

We are actually not interested, eventually, in the average value of y_i^* but in social welfare $W(y_1^*, \dots, y_n^*)$. The social welfare function W adopted here is the constant elasticity of substitution (CES) function:

$$W(y_1^*, \dots, y_n^*) = \left[\frac{1}{n} \sum_{i=1}^n (y_i^*)^{1-\nu} \right]^{\frac{1}{1-\nu}}, \quad (6)$$

where ν is the coefficient of inequality aversion. This function is convenient in particular for international comparisons, since it is invariant with respect to the size of the population (a replication of a given population does not change the value of social welfare). It is also homogeneous of degree one and is measured in the same units as y^* , which makes it easy to interpret. In particular, the value of social welfare with this function is equal to quantity y^{**} , which would give the same social welfare if $y_i^* = y^{**}$ for all i . It can be decomposed as follows:

$$W(y_1^*, \dots, y_n^*) = \left(\frac{1}{n} \sum_{i=1}^n y_i^* \right) (1 - I(y_1^*, \dots, y_n^*)), \quad (7)$$

where $I(y_1^*, \dots, y_n^*)$ is the Kolm–Atkinson inequality index. Social welfare is equal to the average index minus a deduction for inequalities.

In view of the fact that we have little information about the distribution of y_i^* , we approximate this formula by computing the deduction for inequality

¹⁰ Our computations even use non-linear features of the representative agent's preferences, which requires an additional argument. Consider a second-order approximation:

$$\delta_i \approx \sum_k \gamma_{ik} (z_k^* - z_{ik}) + \sum_k \eta_{ik} (z_k^* - z_{ik})^2.$$

Assuming no correlation between η_{ik} and z_{ik} , one has

$$\frac{1}{n} \sum_i \sum_k \eta_{ik} (z_k^* - z_{ik})^2 = \sum_k \left(\frac{1}{n} \sum_i \eta_{ik} \right) \left[\left(z_k^* - \frac{1}{n} \sum_i z_{ik} \right)^2 + \frac{1}{n} \sum_i \left(z_{ik} - \frac{1}{n} \sum_j z_{jk} \right)^2 \right].$$

The last term in this expression is the variance of z_{ik} and is neglected in a representative agent approach. However, forgetting the second-order term altogether would be even worse, which is why we include non-linear corrections in our computations.

on ordinary incomes:

$$\begin{aligned} W(y_1^*, \dots, y_n^*) &\approx \left(\frac{1}{n} \sum_{i=1}^n y_i^* \right) - \left(\frac{1}{n} \sum_{i=1}^n y_i \right) I(y_1, \dots, y_n) \\ &= \left(\frac{1}{n} \sum_{i=1}^n y_i \right) (1 - I(y_1, \dots, y_n)) - \frac{1}{n} \sum_{i=1}^n \delta_i. \end{aligned} \quad (8)$$

III. Computing Equivalent Income

Consider a representative agent whose expected utility is separable in consumption and labor:

$$E \sum_{t=0}^T \beta^t [u(c(t)) - a(\ell(t))], \quad (9)$$

where t is computed in months, T is the maximum lifetime, β is the discount factor, $c(t)$ is the consumption vector with utility $u(\cdot)$, and $\ell(t)$ is the quantity of labor with disutility $a(\cdot)$.

We set out to assess the annual performance of an economy by its impact on a hypothetical agent living permanently under these conditions.¹¹ In this hypothetical context, current income is a good proxy for the present value of lifetime income. In absence of risk, the agent's income and, therefore, consumption expenditure is a constant y equal to the current per capita annual income, and labor is a constant $\ell(t) = \ell$ equal to the current per capita quantity of labor.

Assuming away risks affecting the level of earnings for the employed workers, we focus only on two risks. The risk of death or bad health is described by a function of survival in good health $S(t)$, and the risk of unemployment is described by the probability p of falling unemployed within a month and the probability q of finding a job within a month when one is unemployed. It is assumed that death (or bad health) brings utility to zero, and that unemployment lowers expenditures to τy , where τ is the replacement rate.¹² We assume that unemployment does not affect the quantity of labor because it does not bring real leisure. Moreover, if

¹¹ This hypothetical situation admittedly is not realistic. In order to measure the standard of living of the population living now, it would be preferable to consider the current level of consumption and quality of life as well as the prospects for consumption and quality of life in the future decades. Our more modest approach follows Becker *et al.* (2005).

¹² This amounts to assuming that the agent is not able to smooth consumption when hit by unemployment. We make pessimistic assumptions with respect to the impact of unemployment because, in spite of these assumptions, the results suggest its impact on equivalent income is small.

$S(t) = 1$ until retirement age, there is no uncertainty about the quantity of labor and the constant term of labor can be put out of the expectation operator:

$$E \left(\sum_{t=0}^T \beta^t [u(c(t))] \right) - a(\ell), \quad (10)$$

where the coefficient (depending on the dates of beginning and end of activity) is incorporated into the function $a(\cdot)$ itself.

This agent is submitted to the following living conditions for which we want to compute corrections: consumption prices, labor, risk of unemployment, health and household composition.

Prices

Ideally, one should pick a reference price vector and compute the equivalent income that would bring equal satisfaction under this price vector.

We resort instead to standard indices of purchasing power parity (PPP). These do not have good welfare foundations, but they are easily available and we hope that no major mistake follows from using them.¹³ Therefore, y is measured in PPP USD for all countries.

We retain the same indirect utility function $v(y)$ of PPP expenditures as in Becker *et al.* (2005):

$$v(y) = \frac{1}{1-\varepsilon} y^{1-\varepsilon} + u_0, \quad (11)$$

where $u_0 < 0$ determines a subsistence level y_0 at which utility equals zero. Consistent with the assumption that the agent is not able to smooth consumption when unemployed, henceforth the term $v(y(t))$ replaces $u(c(t))$ in the computation of the agent's lifetime utility.

Labor

Welfare economics suggests a variety of ways to compute equivalent income correcting for labor.¹⁴ We rely here on a particular measure, due to Maniquet (1998), which consists of picking a reference value of labor and computing the agent's WTP to have his labor at this quantity. The advantage of this approach, given data limitations, is that if the reference value of labor is not far from the actual quantity, one can rely on marginal rates of substitution in order to approximate the WTP.

¹³ See, however, Jorgenson (1990) on the error due to using ordinary price indices in the comparison of living standards over time in the US. It would certainly be a serious improvement on current methods to rely on equivalent incomes to compute purchasing parities.

¹⁴ See e.g. Fleurbaey and Maniquet (2005).

The correction term for a quantity of labor different from the norm ℓ^* is then computed by the formula

$$E \left(\sum_{t=0}^T \beta^t [v(y(t))] \right) - a(\ell) = E \left(\sum_{t=0}^T \beta^t [v(y(t) + \delta_\ell)] \right) - a(\ell^*), \quad (12)$$

where δ_ℓ is the value of the correction.

Let us neglect the risks of death and unemployment and suppose that labor is chosen under a budget constraint $y = w\ell + y_0$, where w is the net income of an hour of work and y_0 is a constant. The first-order condition reads

$$\sum_{t=0}^T \beta^t [v'(y)] w = a'(\ell). \quad (13)$$

If the difference between ℓ and ℓ^* is small, one has

$$\delta_\ell \approx \frac{(\ell^* - \ell) a'(\ell)}{\sum_{t=0}^T \beta^t [v'(y)]} = w(\ell^* - \ell), \quad (14)$$

which illustrates how the pricing method can be used in order to estimate the equivalent income.

The norm ℓ^* is the median of our sample of countries. The value of ℓ is computed by assuming that the unemployed (twice the official unemployment rate, to take into account hidden unemployment or discouraged workers) do not have more leisure than the average worker of their country, and that prisoners work twice as much as the average worker (reflecting the strong constraints on their time).

Risk of Unemployment

We now turn to the risk of unemployment. Let d denote the average duration in unemployment (measured in months). The probability q of getting out of unemployment is $q = 1/d$ because $d = 1 + (1 - q) + (1 - q)^2 + \dots = 1/q$.

The probability p of falling into unemployment in the next month can be expressed as

$$p = \frac{r_u}{(1 - r_u)d}, \quad (15)$$

where r_u is the unemployment rate. This is because, when the unemployment pool U and the employment pool E are stable, one has $qU = pE$ and

$$\frac{U}{E} = r_u \frac{U + E}{E} = \frac{r_u}{1 - r_u}. \quad (16)$$

With a fixed employment income y and a replacement rate τ we get (summing from the beginning of activity A up to retirement age R):

$$\begin{aligned} E\left(\sum_{t=A}^R \beta^t [v(y(t))]\right) \\ = \beta^A [pv(\tau y) + (1-p)v(y)] \\ + \beta^{A+1} [(p(1-q) + (1-p)p)v(\tau y) + (pq + (1-p)^2)v(y)] \\ + \beta^{A+2} \left[((p(1-q) + (1-p)p)(1-q) + (pq + (1-p)^2)p)v(\tau y) \right. \\ \left. + ((p(1-q) + (1-p)p)q + (pq + (1-p)^2)(1-p))v(y) \right] + \dots \end{aligned}$$

Let us focus on the factors of $v(\tau y)$. Their sum is equal to

$$\chi = \beta^A \frac{p}{p+q} \left[\frac{1 - \beta^{R-A+1}}{1 - \beta} - (1-p-q) \frac{1 - \beta^{R-A+1}(1-p-q)^{R-A+1}}{1 - \beta(1-p-q)} \right]. \quad (17)$$

We neglect the consequences of unemployment over retirement income. For the computation of the risk premium π , one writes

$$\sum_{t=0}^T \beta^t [v(y - \pi)] = \chi u(\tau y) + \left(\sum_{t=0}^T \beta^t - \chi \right) v(y). \quad (18)$$

Let $\xi = \chi / \sum_{t=0}^T \beta^t$. The above equation simplifies into $v(y - \pi) = \xi v(\tau y) + (1 - \xi)v(y)$.

A standard approximation of the risk premium reads

$$\pi \approx \xi(1 - \tau)y + \frac{1}{2}\xi(1 - \xi) \left(\frac{-v''(y)}{v'(y)} \right) [(1 - \tau)y]^2. \quad (19)$$

We consider that τ is lower than the observed replacement rate, because of the stigma of being unemployed. In other words, the equivalent income loss is greater than the monetary loss. Let $\hat{\tau}$ denote the observed replacement rate.

We can assume that $\xi(1 - \hat{\tau})y$, the expected value of (ordinary) income loss, is already registered in our data of average income. The correction for unemployment risk is therefore the remainder only, namely

$$\delta_U \approx -\xi(\hat{\tau} - \tau)y - \frac{1}{2}\xi(1 - \xi) \left(\frac{-v''(y)}{v'(y)} \right) [(1 - \tau)y]^2. \quad (20)$$

Taking the above constant relative risk aversion (CRRA) function v with coefficient of relative risk aversion $\varepsilon = -v''(y)y/v'(y)$, one finally obtains

$$\delta_U \approx -\xi(\hat{\tau} - \tau)y - \frac{1}{2}\xi(1 - \xi)\varepsilon(1 - \tau)^2y. \quad (21)$$

Health

In the absence of precise data on survival functions at different ages, we consider that $S(t) = 1$ for $t \leq T$, where T is the life expectancy (in good health) at birth, and $S(t) = 0$ for $t > T$. The correction for life expectancy different from the norm T^* then simply amounts to computing

$$E \sum_{t=0}^T \beta^t [v(y(t))] - a(\ell) = E \sum_{t=0}^{T^*} \beta^t [v(y(t) + \delta_S)] - a(\ell). \quad (22)$$

If one ignores the risk of unemployment and assumes a constant y over time, the computation simplifies into $(1 - \beta^{T+1})v(y) = (1 - \beta^{T^*+1})v(y + \delta_S)$, yielding

$$\delta_S = v^{-1} \left(\frac{1 - \beta^{T+1}}{1 - \beta^{T^*+1}} v(y) \right) - y. \quad (23)$$

With the utility function retained here, we therefore have

$$\delta_S = \left[\frac{1 - \beta^{T+1}}{1 - \beta^{T^*+1}} y^{1-\varepsilon} + \frac{\beta^{T^*+1} - \beta^{T+1}}{1 - \beta^{T^*+1}} u_0 (1 - \varepsilon) \right]^{\frac{1}{1-\varepsilon}} - y. \quad (24)$$

For u_0 and ε we retain the same values as Becker *et al.* (2005). The norm T^* is the maximum value in our sample, in view of the idea that the norm should correspond to a notion of good health, as mentioned in Section II.

Household Composition

If the agent is a single, the function $v(y)$ is simply his indirect utility obtained from income y . If he lives in a household with h members, the income y is corrected so that the same function v still measures his indirect utility. The correction is made as follows. We assume that a household spends income in private expenditures hc (equally shared among the h members) and in local public goods g (common rooms, heating, etc.), in fixed proportions. This means that there is an underlying (semi-)direct¹⁵ Cobb–Douglas utility function for a representative member $U(c, g) = \varphi(c^\eta g^{1-\eta})$, which is maximized under the budget constraint $hc + g = y$.

The indirect utility of a member is then equal to

$$\varphi \left(\left(\eta \frac{y}{h} \right)^\eta ((1 - \eta)y)^{1-\eta} \right) = \varphi \left(\eta^\eta (1 - \eta)^{1-\eta} \frac{y}{h^\eta} \right). \quad (25)$$

In order to bring all individuals to a comparable situation, we compute the income that would bring the same indirect utility to the member if

¹⁵ This is not a full-fledged direct utility function, since c and g are still composite commodities.

he were a single.¹⁶ For a single with income $\hat{y}$, the indirect utility equals $\varphi(\eta^\eta(1-\eta)^{1-\eta}\hat{y})$, so the two utility levels are equal if and only if $\hat{y} = y/h^\eta$.

For $\eta = 0.5$, one has to correct the income of individuals by considering their household income divided by the square root of their household size. This corresponds to the OECD equivalence scale factor, which we retain in our benchmark computation. Instead of applying this correction to the representative agent, dividing his income by the square root of average household size, we borrow correction factors from OECD (2005), where sample data are used in order to compute the correction for each household and eventually obtain an average equivalent income. This is better because it takes account of the distribution of household sizes in the population. However, we observe that the two methods provide very similar results. This allows us to make variants on η relatively easily.

Other Possible Corrections

Other corrections could be envisaged but are not carried out here due to lack of data. We discuss below the implications of not making them, in terms of implicit assumptions. We hope that improvement in data availability will make it possible to make some of these additional corrections in the future.

Climatic Conditions. The same income has less value if its primary use is heating or air conditioning. Extreme temperatures are a nuisance that reduce the standard of living. For countries in the same temperate area, which is the case for most countries of our sample, these corrections would be small.

Geostrategic Conditions. The same income has less value if its primary use is national defense instead of consumption. Countries benefiting from the protection of others have a boost in their standard of living. This kind of correction would decrease the level of the US and increase that of Europe. It is, however, difficult to make the distinction between constraints and national preferences. It would be unfair to consider bellicose countries poor if this is the population's choice.

Public Goods (Other than Collective Consumption within Households). Governmental production is registered in GDP at its cost, not in terms of benefit to the population. Countries differ in the size of their public sector

¹⁶ This approach corresponds to what Browning *et al.* (2006) call “indifference scales”, as opposed to “equivalence scales”, which rely on a household utility function and cannot be identified from demand data.

and the level of production of public goods. If the public/private ratio reflects the population's preferences, there is no reason to make a correction for such differences. If it does not accurately reflect the population's preferences, then one should compute the equivalent income that, with an optimal public/private mix for each population, would provide the same satisfaction. The quality of the environment in a country is another kind of public good that contributes to the well-being of its inhabitants. However, WTP for the environment is not easy to estimate, and international differences may be small; environmental problems are roughly the same magnitude in all industrialized countries and/or are more global than local (many pollutants do not stop at borders).¹⁷

Household Production. GDP is often criticized for failing to record non-market production accurately, especially activities usually dominated by women, such as cooking, cleaning and child-rearing. From the perspective of living standards, what matters is determining constrained time, not production. It really makes no difference whether an individual uses his spare time to prepare nice meals or to read philosophy. Insofar as domestic constrained time is approximately the same, per capita, across countries, there is little need to make corrections for this in the computation of average equivalent incomes. However, in the computation of *inequalities* of equivalent incomes, task sharing within households might matter greatly.

Education. In our computation we do not count pupils and students among workers, assuming that educating oneself brings direct benefits that compensate for the effort. But it might be more accurate to seek a direct estimation of the direct benefits of being educated, as distinct from the indirect benefits due to higher productivity (which are registered in GDP figures).

Social Relations. The quality of social relations is a kind of public good that matters a lot for the quality of life. This has to do with security against aggressions, social relations at work, conviviality in one's neighborhood, and the distribution of social status. In view of the likely importance of this dimension for welfare, it appears to be an urgent task to seek ways to take account of it in the computation of equivalent income.

Future Prospects. As explained at the beginning of this section, we have limited our analysis to the hypothetical case of a population permanently exposed to the current conditions. It would be better to take account of the

¹⁷ The World Bank (2006) proposes an evaluation of damages due to particulate emissions (PM10 specifically). It is based on WTP to avoid death caused by such emissions. We do not replicate this, as our correction for health already captures such causes of death or disability.

real prospects of growth and changes in quality of life, which may differ significantly across countries. An even more delicate issue is whether one should also take account of the welfare of future generations. We follow Neumayer (2004), who advocates separating the measurement of the welfare of the current generation from the analysis of the prospects of welfare for future generations, limiting ourselves to the former issue.¹⁸ There is an important body of literature that seeks ways to estimate social welfare over all generations in terms of net national product (NNP),¹⁹ but the definition of social welfare over all present and future generations is a very controversial issue.²⁰

IV. Results

Corrections and Rankings

The main results are presented in the following figure and tables. Details about the data are provided in the Appendix. Figure 3 shows the relative positions of countries after each correction. Table 3 provides the ranking of the 24 countries of our sample after each correction. Finally, values of each correction in dollar terms (per capita) are shown in Table 4.

Considering gross national income instead of gross domestic product has little impact, except for Ireland, which suffers large capital income outflows; and Luxembourg, where the labor force is composed of a large share of non-residents. At the opposite end of the spectrum, Switzerland and France benefit the most, in relative terms, from this first correction.

The correction for leisure is positive for many European countries, particularly France (in spite of a high unemployment rate, since the unemployed are assumed not to enjoy more leisure time than the employed), Norway and the Netherlands. It is neutral as far as the United Kingdom is concerned and negative for Japan, Luxembourg, the United States and Switzerland.

The risk of unemployment affects the relative well-being of countries very little, in spite of our correction for the stigma incurred by the unemployed. Three facts can explain this finding. First, countries with high unemployment often have a long unemployment duration, a low turnover and

¹⁸ In Fleurbaey and Gaulier (2007), we have made simple corrections in terms of genuine savings and in terms of virtual costs of greenhouse gas emissions. The interested reader may consult the impact of such corrections there.

¹⁹ Spun by Weitzman (1976), it is nicely surveyed in Heal and Kriström (2005) and Asheim (2007). It is important to note that this approach only proposes ways to evaluate welfare changes and has not been applied to comparisons across countries with different preferences. For a recent contribution on comparisons between countries (with identical preferences), see Asheim (2008).

²⁰ For a recent collection of contributions to this topic, see Roemer and Suzumura (2007).

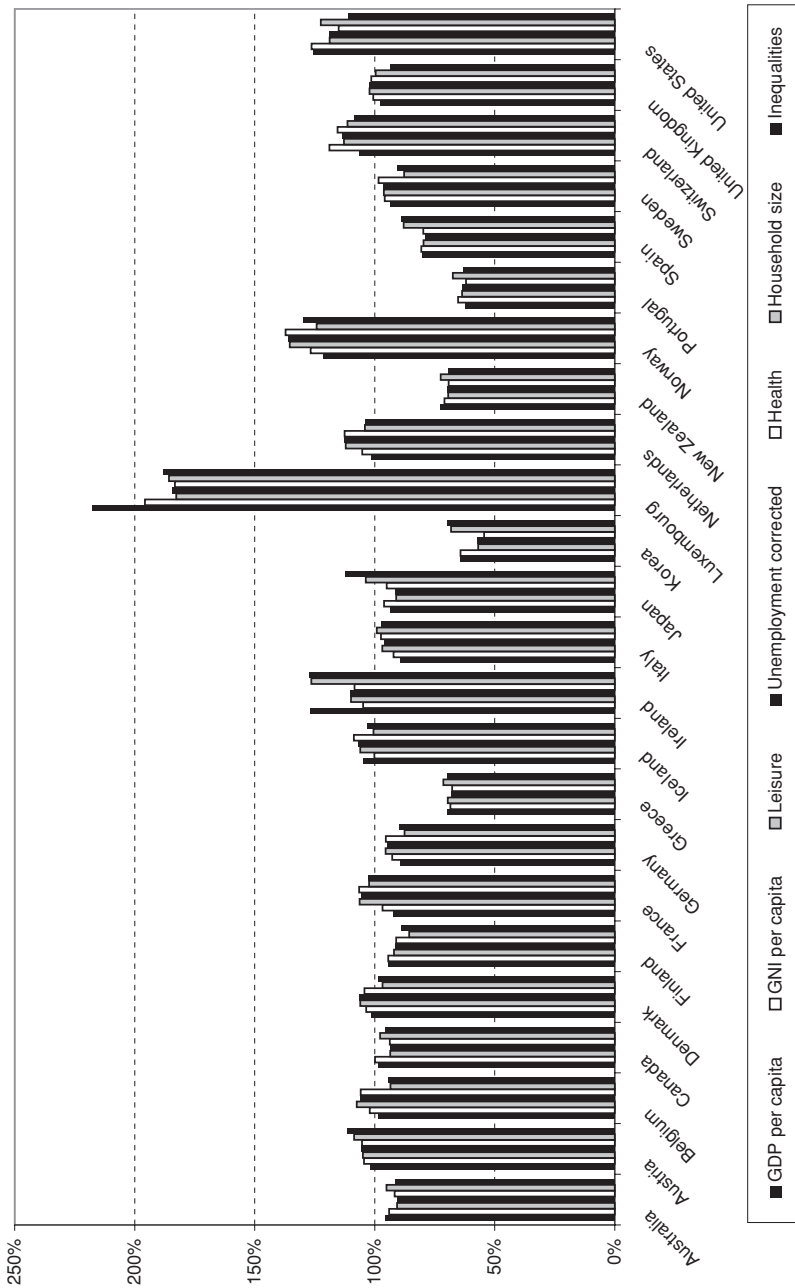


Fig. 3. Relative positions of 24 countries after each correction

Table 2. *GDP per capita and our final indicator*

	GDP per capita	%	Final indicator	%
Australia	30,116	95	33,268	91
Austria	32,176	102	40,513	111
Belgium	31,009	98	34,376	94
Canada	31,129	99	34,838	96
Denmark	31,974	101	35,775	98
Finland	29,816	95	32,371	89
France	29,077	92	37,423	103
Germany	28,147	89	32,712	90
Greece	21,954	70	25,437	70
Iceland	33,090	105	37,597	103
Ireland	40,058	127	46,281	127
Italy	28,162	89	35,388	97
Japan	29,539	94	40,859	112
Korea	20,371	65	25,382	70
Luxembourg	68,719	218	68,438	188
Netherlands	32,056	102	37,892	104
New Zealand	22,912	73	25,204	69
Norway	38,288	121	47,281	130
Portugal	19,687	62	22,957	63
Spain	25,341	80	32,385	89
Sweden	29,499	94	32,939	91
Switzerland	33,541	106	39,522	109
United Kingdom	30,843	98	33,994	93
United States	39,618	126	40,417	111

Note: In levels and relative to the sample average.

therefore a moderate probability of unemployment for employed workers. Second, they also have more generous unemployment benefits, lowering the cost of unemployment. Last, the loss due to the lack of production by unemployment is not included in this correction since it is already subtracted from the (potential) GDP. In this correction we only consider the second-order term of the risk premium (plus the stigma). Total risk premiums, which measure the sacrifice a citizen is willing to accept to suppress the unemployment risk, are far from negligible in some countries, such as France (3.9% of income), Spain (4.6%), Italy (5.2%) and Greece (7.3%).

The correction for health is beneficial for Japan, France, Sweden and Switzerland. On the other hand, Korea and the United States (and to a smaller extent Denmark and Ireland) incur negative corrections.

The correction for household size induces large increases in equivalent incomes, which result from the fact that a single household is taken as reference. For a two-person household the correction is a 41% increase (a couple with a total income of 200 and therefore a per capita income of 100 has the same standard of living as two singles with 141 each). In relative terms, this correction is in favor of southern European countries, Ireland,

Table 3. *Ranking of countries after each cumulative correction*

	GDP per capita	GNI per capita	Leisure	Unemployment	Health	Household size	Inequalities
Australia	13	17	19	19	18	15	16
Austria	7	7	11	11	10	6	5
Belgium	11	9	7	9	9	16	14
Canada	10	12	16	16	17	13	13
Denmark	9	8	10	8	11	14	11
Finland	14	16	17	17	19	20	20
France	17	13	8	10	8	9	10
Germany	19	18	15	15	15	19	18
Greece	22	22	21	22	22	22	21
Iceland	6	11	9	7	6	10	9
Ireland	2	6	6	6	7	2	3
Italy	18	19	13	14	14	12	12
Japan	15	14	18	18	16	8	4
Korea	23	24	24	24	24	23	22
Luxembourg	1	1	1	1	1	1	1
Netherlands	8	5	5	5	5	7	8
New Zealand	21	21	22	21	21	21	23
Norway	4	2	2	2	2	3	2
Portugal	24	23	23	23	23	24	24
Spain	20	20	20	20	20	17	19
Sweden	16	15	14	13	13	18	17
Switzerland	5	4	4	4	3	5	7
United Kingdom	12	10	12	12	12	11	15
United States	3	3	3	3	4	4	6

the United States and Japan. In contrast, northern European countries lose ground.²¹

At this step, we can highlight many switches in the ranking. Obviously, when gaps are small between countries (particularly in the middle of the sample), the ranking can change radically without large variations in the indicator. Countries that lose ground are Denmark (from 9th to 14th position), Finland (14th to 20th) and Australia (13th to 15th). Winners include France (from 17th to 9th position), Italy (18th to 12th) and Japan (15th to 8th). France benefits mainly from a short working time but also from a favorable health performance. Belgium is quite similar to France (high leisure, high productivity, etc.) but suffers from a small household size. Germany, with almost the same GDP per capita as France, benefits less from its short working time and also has a relatively low household size. As a consequence, it gains only one rank, from 19th to 18th.

²¹ The data for some countries, such as Belgium and the Netherlands, are dubious. They have a very low household size in the database we use (OECD) but not with such sources as the United Nations Economic Commission for Europe.

Table 4. *Absolute corrections (in USD per capita)*

	GNI per capita	Leisure	Unemployment	Health	Household size	Inequalities
Australia	-3,226	-1,238	-366	-671	16,319	-7,666
Austria	-2,286	-195	-290	-1,150	18,499	-6,241
Belgium	-1,809	1,225	-788	-1,217	11,827	-5,871
Canada	-2,549	-2,118	-388	-905	16,917	-7,248
Denmark	-2,334	360	-322	-1,673	13,655	-5,885
Finland	-2,806	-989	-455	-1,113	12,416	-4,497
France	-1,397	2,386	-602	-872	15,498	-6,667
Germany	-1,607	476	-532	-889	12,104	-4,987
Greece	-2,354	84	-696	-790	12,560	-5,321
Iceland	-4,440	1,359	-179	-660	14,115	-5,689
Ireland	-10,058	1,099	-307	-1,696	25,347	-8,163
Italy	-1,802	1,048	-641	-628	16,559	-7,310
Japan	-2,039	-1,740	-260	0	19,155	-3,796
Korea	-1,991	-2,275	-133	-1,366	14,747	-3,971
Luxembourg	-12,729	-4,295	-258	-2,262	30,775	-11,512
Netherlands	-1,966	1,617	-280	-1,226	14,635	-6,944
New Zealand	-2,612	-673	-179	-866	12,633	-6,011
Norway	-2,048	2,059	-291	-1,185	16,639	-6,181
Portugal	-1,007	-681	-258	-1,105	12,428	-6,107
Spain	-2,291	-497	-572	-562	16,458	-5,491
Sweden	-2,089	-200	-313	-482	11,361	-4,837
Switzerland	479	-2,094	-263	-654	16,949	-8,437
United Kingdom	-2,043	109	-322	-1,359	15,654	-8,888
United States	-3,488	-2,515	-432	-2,306	21,858	-12,318

Rankings are striking, but it is better to examine relative levels as shown in Figure 3 and Table 2. Denmark and the US lose nearly four points of average GDP equivalent each, but this fall leads to a five-rank loss in the case of Denmark compared with only one position lost by the US. France, Italy and Japan gain nearly 10 points.

Let us now turn to the last correction. The correction for inequalities is strong, in spite of our conservative choice for the inequality-aversion coefficient. Ireland, the UK, the US, Australia, Portugal and Italy retrograde. At the opposite end, the correction is beneficial to Japan and the northern European countries. It is neutral for the relative position of France, Canada, the Netherlands and Spain.

Finally, Luxembourg loses 30 points (with respect to the sample average) but remains first. Norway (+9 points) fills the gap and ends up second. The US loses 15 points and ends up only 11 points above the average. Finland and the UK lose about 6 and 5 points, respectively. At the opposite end, the main winners are Japan (+18 points) and France (+11 points), followed by Spain, Austria, Italy and Norway (about 9 points each).

The results show that the corrections are far from negligible. Not surprisingly, the final index of living standards is still correlated with GDP per capita, but the general ranking of countries is substantially affected by the corrections. The high correlation between our final indicator and GDP per capita (96% for levels, 86% for ranks) should not hide that none of our corrections is significantly correlated with GDP per capita (the correction for unemployment is positively correlated with GDP, but the correlation is only 42%). The most important corrections (health, leisure, household size and inequalities) appear to be independent of GDP, which may come as a surprise concerning the health performance.

Three Welfare Patterns

Several groups of countries with similar non-income features can be identified and associated with different models of social and economic development. Anglo-Saxon countries share many characteristics. The US, the UK, Canada, Australia and New Zealand generally have (i) high inequalities, (ii) a high working-time (except the UK) and (iii) a low cost of unemployment but a high turnover on the labor market (except Australia). This Anglo-Saxon model loses its homogeneity when it comes to health, since Australia and Canada have an above-average performance. France, Italy and Spain can be grouped together in a “Latin” pattern. They share (i) a high cost of unemployment, (ii) a high level of leisure and (iii) a high life expectancy. Portugal is left out of this “Latin” group. It did not catch up with the richest European countries as Spain did, and it has a relatively poor health performance as well as a working-time above the average. Belgium, and to a lesser extent Germany, could be included in the “Latin” group but, with low household sizes and low inequalities, these countries are close to another pattern that we label as “Nordic”. This last group gathers Norway, Sweden, Denmark and Iceland, but also Austria and the Netherlands. These rich countries are among the most egalitarian, have a low unemployment risk, and a low working-time. Several other European countries are close to this model but exhibit at least one significant divergence: Belgium and Germany as noted before (with high unemployment), Switzerland and Finland (with high working-time).

Sensitivity to Parameters

The computation of equivalent incomes is sensitive to estimated parameters of preferences, to the choice of reference values for quality of life, and to the ethical parameter of aversion to inequality. In the Appendix, we propose a few variants that are constructed to be favorable to each one of the three groups of countries identified above.

V. Conclusion

The purpose of this work is not to provide definitive figures about living standards, but to introduce a method and illustrate the kind of results it can yield. It seems to us that an approach of this kind is superior to composite social indicators, such as the HDI, because it is grounded in economic theory and in particular is meant to take account of the population preferences about the various dimensions of life. In contrast, composite indicators are commonly criticized as involving weights for which no rational basis of evaluation can be proposed. Another problematic feature of such indicators is that they aggregate sub-indices per domain and do not take account of the accumulation of disadvantages in all dimensions among the poorest. In contrast, equivalent incomes should ideally be computed at the individual level before social aggregation is performed. We have not been able to make much of this feature of the approach because of data limitations, but better data would make a difference. Another limitation of our work is that it uses annual data, computing the well-being of a hypothetical population permanently exposed to these conditions. Obviously, the well-being of members of the present generation also depends on their situation in recent decades and on their future prospects. Again, incorporating this would be much more demanding in terms of data.

At the very least, our computations show that usual comparisons in terms of GDP per capita are very fragile. It is often asserted that GDP is the most robust figure one can use for international comparisons. Specialists of national accounts, however, know that GDP figures themselves rely on many approximations, even before PPP corrections are made. But the main problem with this kind of judgment is the implicit assumption that it is better to be precisely wrong than to be approximately right.

Our tables start from GDP and introduce a series of corrections. Such a full display of corrections seems advisory. Those who only trust the column of GDP figures can still look at it exclusively, but there is no reason to censor the corrections for labor, unemployment risk, health, and so on. Moreover, our corrections are not based on arbitrary weights that reflect ethical preferences (except the correction for inequalities, which is based on a coefficient of inequality aversion). They are based on WTP figures that can be discussed on an empirical basis and can (we hope) be made more accurate in the future.

For an accurate application of this methodology, one needs survey data on income and on the additional dimensions of consumption that have been listed here, as well as on preferences (WTP), at the individual level and for all the countries studied. WTP data can be collected using contingent valuation questionnaires or indirect methods based on revealed preferences or on satisfaction surveys. The key to making real progress is to collect

data at the individual level in order to avoid the approximations implied by relying on average data and a representative-agent approach. We hope that our work will encourage the production of such data. In the meantime, we hope it also shows that even with the available data it is possible to obtain some indication of how international comparisons should be revised when additional dimensions of standards of living are taken into account.

Appendix

Data

- GDP, GNI per capita: World Bank, International Comparison Program database.
- Hours of work, Unemployment rate, Working population, Duration of unemployment, Replacement rates: OECD, Labor Statistics, and OECD, Benefits and Wages. For some countries, duration of unemployment estimated from share of long-term unemployment (duration = $35.35 * \text{share of long term} + 2.0408$, $R^2 = 0.87$).
- Wage shares: OECD–STAN.
- Prisoners: International comparisons of criminal justice statistics 2000 by G. C. Barclay and C. Tavares, updated 30 May 2002, International Center for Prison Studies (www.kcl.ac.uk/depsta/rel/icps/) for Iceland, Korea and Norway.
- Taxes: Revenue Statistics, 1965–2003, OECD, Paris, 2004; Taxing Wages, 2003–2004, OECD, Paris, 2004; OECD Tax Database.
- Health-adjusted life expectancy: OECD Health Data 2005, October 2005.
- Household size: OECD, 2002, except Iceland, 1995: Euromonitor—European Marketing Data and Statistics 1997; 32nd edition; Korea, 2000: “Household Projections for the Republic of Korea”, Y.-J. Park, H.-S. Kim and H.-J. Ko, paper for 20th Population Census Conference, 19–21 June 2002, Ulaanbaatar, Mongolia; Ireland and Luxembourg, 2001: United Nations Economic Commission for Europe.
- Distribution of income: World Bank and Luxembourg Income Study database. Various years between 1990 (Spain) and 2000 (Finland, Germany, Italy, Luxembourg, Norway, Sweden and the United States).
- *Parameters*: In the benchmark computation:
 - the annual discount factor ρ is fixed to 0.03.
 - the stigma of being unemployed translates into replacement rates 20 percentage points below the observed replacement rate, i.e. $\tau = \hat{\tau} - 0.2$.
 - the coefficient of risk aversion ε is 0.8.
 - the coefficient in the household size correction is $\eta = 0.5$.
 - following Becker *et al.* (2005), we set the elasticity of the instantaneous utility function to 0.346 (as estimated by Murphy and Topel, 2003, using US data), which together with the intertemporal elasticity of substitution (1.25 or the inverse of our coefficient of risk aversion) gives a “subsistence” daily income of

about 1 USD. In the words of Becker *et al.*, an individual with this income would be indifferent between being alive or dead.

— the coefficient of social preference for equality is $\nu = 1.5$.

Sensitivity Analysis

One can be confident with respect to the direction of the corrections since that depends mainly on the relative situations of countries in terms of relatively easily comparable data, such as time at work, life expectancy, household size and inequalities. On the contrary, the extent of the corrections can vary greatly with hypotheses and choice of parameters. The results appear the most sensible to assumptions concerning preferences about working time, health and household size. The parameter for inequalities is also very important, but this is an ethical parameter that can be changed at will by the user of the data.

Table A1 presents three alternative computations. Besides our benchmark scenario, three alternative sets of parameters are considered. One is especially favorable to Anglo-Saxon countries, with low WTP for leisure (3/4 the value taken in the benchmark) and

Table A1. *Alternative sets of parameters*

	GDP per capita (%)	Benchmark (%)	Anglo-Saxon (low WTP for leisure; low aversion for inequalities) (%)	Latin (high WTP for leisure; high risk aversion; high subsistence level) (%)	Nordic (low share of collective consumption within households; high inequality aversion) (%)
Australia	95	91	97	91	80
Austria	102	111	110	112	113
Belgium	98	94	93	95	102
Canada	99	96	98	95	89
Denmark	101	98	98	99	100
Finland	95	89	87	89	100
France	92	103	101	105	106
Germany	89	90	88	91	98
Greece	70	70	70	70	68
Iceland	105	103	101	105	108
Ireland	127	127	124	128	120
Italy	89	97	98	98	92
Japan	94	112	108	112	122
Korea	65	70	70	70	62
Luxembourg	218	188	187	180	195
Netherlands	102	104	104	106	106
New Zealand	73	69	71	70	63
Norway	121	130	124	131	151
Portugal	62	63	66	64	54
Spain	80	89	88	89	85
Sweden	94	91	88	91	103
Switzerland	106	109	112	108	106
United Kingdom	98	93	97	94	86
United States	126	111	119	108	91

low aversion to inequalities (0.5 instead of 1.5). Another takes a different direction and turns out to favor Nordic countries especially, with a low share of collective consumption within households (0.25 instead of 0.5) and a higher aversion to inequalities (3 instead of 1.5). A third one is more favorable to Latin countries, with a higher WTP for leisure (1/4 above the benchmark) and a higher risk aversion (2 instead of 0.8).²²

The sensitivity of the results to the set of assumptions is obvious. There are, however, some robust features. For instance, over all variants, the US loses at least 7 points with regard to its relative position to the mean in terms of GDP, whereas Japan gains at least 14 points and France gains 9 points. There are countries whose relative position remains rather stable across the variants, such as Austria, the Netherlands, Italy, Denmark, Spain and Greece.

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²² A greater risk aversion increases the correction for unemployment but also the correction for health, and these countries gain more from the latter change than they lose from the former. A higher risk aversion also induces a greater value for the subsistence level (about 18 USD per day).

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